



Madhav Gupta
Electrical Engineering
Indian Institute of Technology Bombay
Specialization: Communication & Signal Processing

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Dual Degree (B.Tech. + M.Tech.)
Gender: Male
DOB: 06/08/2003

Examination	University	Institute	Year	CPI / % Credits
Graduation	IIT Bombay	IIT Bombay	2026	

Pursuing Minor degree in **Artificial Intelligence and Data Science** offered by CMInDS, IIT Bombay

SCHOLASTIC ACHIEVEMENTS

- Achieved **99.989 percentile** and All India Rank **164** in **JEE Main** examination amongst 1 million+ students [2021]
- Secured All India Rank **724** out of 0.15 million+ candidates who appeared for **JEE Advanced** examination [2021]
- Recipient of Kishore Vaigyanik Protsahan Yojana (**KVPY**) Fellowship organised by **IISc Bangalore** [2021]
- Recipient of the prestigious National Talent Search Examination (**NTSE**) scholarship, ranked **3rd** in Stage-1 [2019]
- Qualified for Regional Maths Olympiad(**RMO**) by securing a rank in **Top 1%ile** in Pre-RMO by HBCSE [2018]

PROFESSIONAL EXPERIENCE

Software & AI Solutions Intern | Hilti Group

[May '24 - Dec '24]

Awarded a **Pre-Placement Interview** for full-time return offer

- Engineered an **employee management system** using Angular, .NET APIs, & MongoDB with robust **CRUD** operations
- Developed bulk Bill of Materials export in channel calculator app using **microservices**, achieving **5x** faster export
- Integrated **4** apps via APIs, SharePoint, & **Copilot Studio**, boosting info search efficiency by **3x** across platforms
- Presented a successful **POC** to leadership, securing approval for future scaled product development for **10K+** users

Full-Stack AI Intern | NuSummit (National Stock Exchange-IT)

[May '25 - Aug '25]

- Built a POC **RAG-based Q&A** system for financial documents using **Ollama** & **Qdrant**, handling PDFs of 10+ pages
- Integrated end-to-end pipeline using **FastAPI** with user auth and enhanced Angular UI for secure **PDF querying**
- Explored **RAGAS** framework to implement guardrails and enhance **context matching** for accurate response evaluation

AI Research Intern | Sentry AI

[Jan '25 - May '25]

Awarded a **Letter of recommendation** for outstanding work on real-world computer vision problems

- Boosted loitering detection by **50%** via **contrastive learning** for background-invariant clustering across 50+ cameras
- Fine-tuned a Swin Transformer with **K-fold cross-validation** to segregate suitable images, achieving **83%** accuracy
- Slashed false positives by **80%** across **100+** cameras using height-vs-(X,Y) coordinate filtering via **2D linear regression**

Conversational AI and LLM Intern | SameSpace

[Nov '23 - Dec '23]

- Built end-to-end **conversational analytics pipeline** spanning data preparation, translation, **fine-tuning**, and inference
- Created **200+** conversations with their **pain points**, agent scores, semantics to fine-tune GPT-2 for their generation
- Translated **OpenOrca** and Samsun datasets using **Opus** into Tagalog, fine-tuned a **GPT-2 summarizer** model on them using the **SimpleTransformers** library to generate concise summaries of customer conversations in Tagalish

PUBLICATIONS

- Kavitha Viswanathan, **Madhav Gupta**, Amit Sethi et al., "**FANVID: A Benchmark for Face and License Plate Recognition in Low-Resolution Videos**", submitted to **NeurIPS 2025**(Datasets and Benchmarks Track, under review)

TECHNICAL & RESEARCH EXPERIENCE

IIT Bombay Racing - Driverless

Part of a 70+ member team designing driverless electric race cars for Formula Student competitions in Germany/UK

Formula Student AI 2023 | Representing the Team at Silverstone

[May '23 - Jul '23]

- Part of the **First ever contingent** of the only Indian team to participate in FS-AI 2023 at the Silverstone Circuit
- Engineered **CAN**(Controller Area Network) interface between **AI** unit and vehicle control unit of **Driverless Car**
- Delivered a **Business Plan Presentation** for an advanced **Level 4 autonomous system** to international judges

Robust Visual Attribution via Submodular Optimization for OOD Data

[Jan '25 - Present]

Thesis Project | Guide: Prof. Ganesh Ramakrishnan

- Built a submodular selection and **perturbation-based uncertainty** framework for enhanced vision model interpretability
- Achieved higher Insertion AUC(**0.91** on synthetic OOD) over prior in-distribution submodular models with **ResNet-101**
- Drafting a research paper for **IEEE WACV 2025** on robust visual attribution for **Out-of-Distribution** scenarios

Network Inversion for Enhanced NLP Model Interpretability

[Jan '25 - May '25]

R&D Project | Guide: Prof. Amit Sethi

- Designed **Bi-LSTM** and **transformer-based framework** to reconstruct input sequences, boosting model interpretability
- Achieved **93.9%** accuracy in sentiment classification & 63.2% in **sentiment-guided** text generation on Twitter dataset
- Developed custom transformer for **previous token prediction** in LLMs, trained using next token predictor (GPT-2)

Optimizing Object Detection with Detection Transformers (DETR)

[Aug '24 - Nov '24]

Supervised Research Project | Guide: Prof. Amit Sethi

- Evaluated **DETR-based detection**, optimizing backbones (MobileNet-V3, ConvNext, ResNet) on **Pascal VOC** dataset
- Achieved top **mAP (51.4 @50%)** with ConvNext; identified **MobileNet-V3** as optimal for accuracy-efficiency balance
- Fine-tuned encoder-decoder layers and batch sizes to **reduce parameters**, boost mAP, and **mitigate overfitting**

KEY PROJECTS

AI-Powered Tender Discovery and Retrieval System | Industrial Project

[Jan '25 - Mar '25]

Convolution Engineering Consultancy | Transforming tender management using AI

- Built semantic search using **Pinecone**, E5 embeddings and **Reranker** for vector-based ranking across **10K+** tenders
- Implemented **full lifecycle automation**, handling **1K+** new tenders daily, expiry-based deletion and real-time downloads
- Automated **tender scraping** across government portals using **Selenium** generating JSONs with key metadata entities

Multi-Agent AI Engineer for Business Workflows | Industrial Project

[Oct '24 - Nov '24]

Theseus Technologies | Cutting-edge AI solutions for the HR Industry

- Designed a multi-agent system with **AutoGen** to generate **AI workflows** from **business-specific** inputs and I/O specs
- Orchestrated **subtasks via a unified agent** that dynamically spawns or assigns AI agents based on **10+ tool** integrations

AI Moderated Usability Testing | Industrial Project

[Jan '24 - Apr '24]

Userology | Conversational AI to moderate usability testing sessions

- Engineered end-to-end pipeline for usability test evaluation based on **observed patterns**, **categories**, and **policies**
- Built an AI moderator **prompt evaluation** pipeline enabling **3×** faster iteration cycles across **5+** policy dimensions
- Created prompts for **automated summaries**, key moments, important user quotes, reducing analysis time by **70%**

AI Powered Resume Assistant | Course Project

[Oct '23 - Nov '23]

EE782 - Advanced Machine Learning | Guide: Prof. Amit Sethi

- Fine-tuned a **Mistral-7B** model on the scraped IITB resume repository to capture **domain-specific** writing styles
- Utilized **4-bit quantized** and **sharded models** for parallel inference to optimize GPU usage and fine-tune efficiently

AI/ML-Driven Grievance Resolution System | Self Project

[Jan '24 - Feb '24]

DARPG Hackathon 2024 | Organized by Government of India

- Developed an AI solution to auto-categorize **175k grievances** and identify officers for efficient **complaint resolution**
- Extracted unique **complaint flow** sequences and built a **reverse-matching** algorithm using **final remark similarities**
- Built a **TF-IDF vectorizer** to extract features from complaint data, applied **XGBoost** for resolution officer prediction

TECHNICAL PROFICIENCY AND RELEVANT COURSES

- Programming Languages** : Python, JavaScript, TypeScript, C++, C#, SQL, bash
- Frameworks** : Git, Linux, Angular, .NET Core, Specflow, Selenium, MongoDB, CI/CD pipelines, Copilot Studio
- Libraries** : NumPy, Pandas, Matplotlib, PyTorch, TensorFlow, SciPy, Scikit-Learn, OpenCV, NLTK
- Key Courses** : Data Structure and Algorithms, Advanced Machine Learning, Optimization in Machine Learning, Markov Chains, Probability and Random Processes, Database and Information Systems(Ongoing)

POSITION OF RESPONSIBILITY

Interview Coordinator | Placement Office, IIT Bombay

[Aug '22 - Dec '22]

- Coordinated with a team of **250+** members to interview 2000+ students during the **placement season** of 2022-23
- Assisted in conducting Interviews for **20+** firms, handling **student queries**, and maintaining interview records

Event Coordinator | Impulse, EESA's Annual Fest

[Mar '23]

- Acted as a **point of contact (POC)** for top management from companies like **Samsung, Mercedes & Reliance Jio**
- Got a chance to interact with them, fostering **valuable connections** and **enhancing their bonding** with our institution

EXTRA CURRICULAR ACTIVITIES

- Secured **2nd** position in **QuantHive's Algoswarm, Algo-Trading Hackathon**,organised by Finance Club [Aug '23]
- Contributed as a member of the **10-person team** for **DevRev Agent challenge** at **Inter IIT Tech Meet 12.0**[Dec '23]
- Ranked among **top 15** of **250+** teams in Finsearch, presented to **industry professionals** & won cash prize [Aug '22]
- Mentored **10+** participants in **WiDS Bootcamp** on developing a **Sudoku Solver** using **Computer Vision** [Dec '23]
- Achieved **3rd place** in the Electrical Engineering department-wide **table tennis** tournament [Aug '24]